Introduction to

Microsoft Excel

Chapter 1

Excel Interface Overview

Welcome to Microsoft Excel! This chapter introduces you to what Excel is, what it looks like when you open it, and the key elements on the screen. By the end of this chapter, you will be comfortable identifying every part of the Excel window and understanding its purpose.

1.1 What is Microsoft Excel?

Microsoft Excel	A spreadsheet software developed by Microsoft that allows you to organize, calculate, analyze, and visualize data in a grid of rows and columns. It is part of the Microsoft Office / Microsoft 365 suite.

Think of Microsoft Excel as a very powerful digital notebook with superpowers. Instead of just writing down numbers on paper, Excel lets you perform calculations automatically, create charts, sort information, and much more.

🔎 **Real-World Analogy**

Imagine you run a small stationery shop. Every day, you note down which items were sold, how many units, and the price. At the end of the month, you want to know:

- Total sales for each product
- Which product sold the most
- Your total monthly revenue

Doing this by hand is slow and error-prone. Excel does all of this in seconds — automatically. You simply enter the data, and Excel handles all the calculations.

1.2 Opening Microsoft Excel

To open Excel, follow these steps depending on your operating system:

Step	Windows	macOS
1	Click the Start Menu (Windows icon in the taskbar)	Click the Launchpad or open Finder
2	Type Excel in the search bar	Search for Microsoft Excel
3	Click Microsoft Excel from the results	Click the Excel icon to open it
4	Excel opens with a Start Screen	Excel opens with a Start Screen

📝 **Note**

If Excel is not installed on your computer, you can access a free version through your browser at office.com using a Microsoft account. The browser version works similarly to the desktop version.

1.3 The Excel Window — Key Components

When you open a new workbook in Excel, you will see a window divided into several distinct areas. Each area has a specific purpose. The diagram below labels all major parts of the Excel interface.

Component	Location on Screen	What It Does
Title Bar	Very top of the window	Shows the name of the file currently open (e.g., Book1)
Quick Access Toolbar	Top-left corner	Provides shortcuts for frequently used commands like Save, Undo, Redo
Ribbon	Below the Title Bar	Contains all Excel commands organized into tabs and groups
Name Box	Left side, below the Ribbon	Displays the address of the currently selected cell (e.g., A1)
Formula Bar	Right of the Name Box	Shows and allows editing of the content or formula in the active cell
Column Headers	Grey letters across the top (A, B, C...)	Identify each column in the worksheet
Row Headers	Grey numbers along the left side (1, 2, 3...)	Identify each row in the worksheet

Active Cell	Highlighted cell in the grid	The cell currently selected, where your data will be entered
Worksheet Area	Main central grid	The working area where you enter and manage data
Sheet Tabs	Bottom of the window	Show the names of sheets in the workbook (Sheet1, Sheet2...)
Status Bar	Very bottom of the window	Displays useful information like SUM, AVERAGE, COUNT of selected cells
Scroll Bars	Right and bottom edges	Allow you to scroll through the worksheet horizontally and vertically

1.4 Understanding Cells, Rows, and Columns

The most fundamental concept in Excel is the cell. Every piece of data you work with lives inside a cell.

Cell	The smallest unit in an Excel spreadsheet. Each cell is the intersection of a column and a row. A cell can hold text, numbers, dates, or formulas.

Term	Description	Example
Column	A vertical line of cells. Identified by letters.	Column A, Column B, Column C...
Row	A horizontal line of cells. Identified by numbers.	Row 1, Row 2, Row 3...
Cell	The box at the intersection of a column and a row.	Cell A1, Cell B3, Cell D10
Cell Address	The unique identifier of a cell, written as Column Letter + Row Number.	A1, C5, G12
Active Cell	The cell currently selected. Highlighted with a green border.	Whichever cell you click on

🔎 **Analogy — A Grid Like a Timetable**

Think of the Excel grid like your school timetable:

- Columns are like the days of the week (Monday, Tuesday, Wednesday...)
- Rows are like the time periods (Period 1, Period 2, Period 3...)
- Each cell is like a specific slot — for example, Monday, Period 2 = Mathematics

In Excel: Column A, Row 2 = Cell A2. It is an exact, unique location in the grid.

Chapter 2

Navigating the Ribbon

The Ribbon is the command centre of Excel. Almost every action you can perform in Excel is accessible through the Ribbon. This chapter explains how the Ribbon is organized, what each tab contains, and how to find the tools you need quickly.

2.1 What is the Ribbon?

The Ribbon	A panel of organized buttons and menus that runs across the top of the Excel window, just below the Title Bar. It groups all of Excel's features into logical tabs and sections for easy access.

2.2 Ribbon Tabs — Overview

The Ribbon is divided into Tabs. Each tab contains commands related to a specific category of tasks. Here is an overview of the main tabs you will encounter:

Tab Name	What It Contains	When You Use It
File	Save, Open, Print, Share, Options, Account settings	To manage files, print documents, or adjust settings
Home	Basic formatting, clipboard, font, alignment, number format	Most commonly used — for everyday formatting and editing
Insert	Charts, tables, images, shapes, links, text boxes	When adding visual elements or objects to your worksheet
Page Layout	Margins, orientation, paper size, gridlines, print area	When preparing a worksheet for printing or presentation
Formulas	Insert functions, function library, formula auditing	When working with calculations and functions
Data	Sort, Filter, Data Validation, Remove Duplicates, Import Data	When organizing and analyzing data
Review	Spell check, comments, track changes, protect sheet	When reviewing or sharing work with others
View	Zoom, freeze panes, split windows, page break preview	When adjusting how you see the worksheet

2.3 Groups Within a Tab

Within each tab, commands are further grouped into sections called Groups. Each group contains related buttons and controls. For example, the Home tab contains these groups:

Group Name	Commands Inside	Example Use
Clipboard	Cut, Copy, Paste, Format Painter	Copy data from one cell and paste it in another
Font	Font name, size, bold, italic, underline, border, fill color	Change text to bold or make a cell background yellow
Alignment	Horizontal/vertical align, wrap text, merge cells, indent	Center text in a cell or merge two cells into one
Number	Number format dropdown (General, Currency, Percentage, Date...)	Format a number as currency: ₹1,500.00
Styles	Conditional Formatting, Format as Table, Cell Styles	Highlight cells with values above a threshold
Cells	Insert, Delete, Format rows, columns, sheets	Insert a new column between A and B
Editing	AutoSum, Fill, Clear, Sort & Filter, Find & Select	Quickly sum a column of numbers using AutoSum

2.4 Contextual Tabs

Some Ribbon tabs only appear when you select a specific type of object. These are called Contextual Tabs. They provide commands specific to whatever is selected.

When You Select...	Contextual Tab That Appears	Commands It Contains
A chart	Chart Design / Format	Change chart type, add labels, modify chart colours
An image or picture	Picture Format	Crop, adjust brightness, apply artistic effects
A table (formatted)	Table Design	Change table style, add total row, rename table

💡 **Pro Tip**

If you cannot find a button you are looking for, check if you are on the right tab. For example, the Sort & Filter button is in the Data tab, not the Home tab (though a shortcut exists in Home > Editing as well).

2.5 Collapsing and Expanding the Ribbon

Sometimes you may want more space to work on your spreadsheet. You can temporarily hide the Ribbon:

Action	How To Do It
Collapse the Ribbon	Double-click any active tab name, OR press Ctrl + F1
Expand the Ribbon	Click any tab name once, OR press Ctrl + F1 again
Pin the Ribbon open	Click the Pin icon (📌) that appears at the bottom-right of the Ribbon when collapsed

Chapter 3

Understanding Worksheets and Workbooks

Before you start entering data, it is important to understand how Excel organizes its files. Excel uses two key concepts: Workbooks and Worksheets. This chapter explains the difference and how to work with both.

3.1 Workbook vs. Worksheet

Term	Definition	Real-World Analogy
Workbook	The entire Excel file. One workbook is one .xlsx file saved on your computer.	Think of it as a physical binder or folder.
Worksheet (Sheet)	A single page within a workbook. One workbook can contain many worksheets.	Think of each worksheet as one page inside the binder.

🔎 **Analogy — Workbook and Worksheets**

Imagine a company's annual accounts binder:

- The binder itself = the Workbook (the Excel file)
- Page 1 inside the binder = Worksheet named 'January Sales'
- Page 2 inside the binder = Worksheet named 'February Sales'
- Page 3 inside the binder = Worksheet named 'Annual Summary'

All three pages are part of the same binder. Similarly, all worksheets belong to the same workbook.

3.2 Sheet Tabs — Navigating Between Worksheets

At the bottom of the Excel window, you will see Sheet Tabs. Each tab represents one worksheet. You can click on a tab to switch to that sheet.

Action	How To Do It
Switch to a different sheet	Click on its Sheet Tab at the bottom
Add a new worksheet	Click the + (plus) button to the right of the existing tabs
Rename a worksheet	Double-click the tab name, type the new name, and press Enter
Delete a worksheet	Right-click the tab > Select Delete

Move a worksheet	Click and drag the tab to a new position along the tab bar
Copy a worksheet	Right-click the tab > Move or Copy > Check 'Create a copy'
Change tab color	Right-click the tab > Tab Color > Choose a color

📝 **Note**

You can have up to 255 worksheets in a single workbook. However, for good organization, it is recommended to keep the number reasonable and name each sheet clearly (e.g., 'Jan Sales', 'Feb Sales', 'Expenses') so that the workbook is easy to navigate.

3.3 Worksheet Grid Dimensions

Each Excel worksheet contains a very large grid. Understanding its size helps you appreciate how much data a single sheet can handle:

Property	Value
Total Columns per Sheet	16,384 columns (Column A through Column XFD)
Total Rows per Sheet	1,048,576 rows
Total Cells per Sheet	Over 17 billion cells
Last Cell Address	XFD1048576

💡 **Pro Tip**

Press Ctrl + End on your keyboard to jump to the last used cell in your worksheet. Press Ctrl + Home to return to cell A1 instantly. This is very useful when working with large datasets.

Chapter 4

Creating and Saving Workbooks

Now that you understand what a workbook is, this chapter will walk you through how to create one from scratch, use a template, and save it correctly. Saving is one of the most important habits to develop early.

4.1 Creating a New Workbook

You can create a new workbook in three ways:

Method	Steps
From the Start Screen	When Excel opens, click 'Blank Workbook' on the Start Screen
Using a Keyboard Shortcut	Press Ctrl + N (Windows) or Command + N (macOS) at any time
From the File Menu	Click File > New > Blank Workbook

4.2 Using Templates

Excel provides ready-made templates for common tasks. A template is a pre-designed workbook that already has headings, formatting, and sometimes formulas built in. You simply fill in your own data.

🔎 **Analogy — Using a Template**

Think of a template like a printed form from a bank. The form already has all the fields labelled (Name, Account Number, Amount, Signature). You just fill in your details.

Similarly, an Excel Invoice Template already has the company name field, item column, quantity, price, and total formula set up. You just add your data.

To use a template: Click File > New > search for a template name (e.g., Budget) > select a template > click Create. Excel will open a pre-designed workbook ready for your data.

4.3 Saving a Workbook

Saving your workbook ensures your work is not lost. There are several ways to save:

Save Method	Shortcut	When to Use
Save (overwrite existing file)	Ctrl + S	When updating a file you have already saved before
Save As (save with a new name or location)	F12 or Ctrl + Shift + S	When saving a file for the first time or saving a copy
Auto-Save (Microsoft 365 only)	Toggle in top-left corner	For files stored on OneDrive — saves automatically as you type

When saving for the first time using Ctrl + S, Excel will automatically open the Save As dialog. You will need to:

- Choose a location — such as your Desktop, Documents folder, or a specific project folder
- Enter a file name — give it a clear, meaningful name (e.g., 'Monthly_Sales_April_2025')
- Select the file format — for most purposes, keep it as Excel Workbook (.xlsx)

> 💡 **Pro Tip — Save Early, Save Often**
> Get into the habit of pressing Ctrl + S every few minutes while working. Even with Auto-Save, manual saving is a good safety habit. A power cut, software crash, or accidental close can cause you to lose unsaved work.

Chapter 5

Data Entry and Editing

Entering and editing data is the most fundamental skill in Excel. This chapter teaches you how to type data into cells, edit it, move between cells efficiently, and make changes without losing your original data.

5.1 Entering Data into a Cell

To enter data into a cell, simply click on it and start typing. When you are done, you can confirm your entry by pressing one of the following keys:

Key	What It Does After Entering Data
Enter	Confirms the entry and moves the cursor to the cell below
Tab	Confirms the entry and moves the cursor to the cell to the right
Arrow Keys (← ↑ → ↓)	Confirms the entry and moves the cursor in the direction pressed
Escape (Esc)	Cancels the entry and restores the original cell content

5.2 Editing Existing Cell Content

If you need to change the content of a cell that already has data, you have two options:

Method	How To	Best For
Replace entirely	Click the cell and just start typing. The old content is replaced.	When you want to enter completely new content
Edit in-place	Double-click the cell (or press F2) to enter edit mode. Use arrow keys and Backspace to modify.	When you want to change part of the content
Edit via Formula Bar	Click the cell, then click inside the Formula Bar and make changes there.	When the cell contains a long formula or long text

5.3 Selecting Multiple Cells

Before you can format, move, or delete a group of cells, you need to select them. Here are the most common selection techniques:

What to Select	How To Do It

A single cell	Click on it
A range of cells	Click the first cell, hold Shift, click the last cell
An entire column	Click the column letter at the top (e.g., click 'C' to select all of Column C)
An entire row	Click the row number on the left (e.g., click '5' to select all of Row 5)
Multiple non-adjacent cells	Hold Ctrl and click each cell individually
All cells in the sheet	Press Ctrl + A, or click the grey square in the top-left corner (above row 1, left of column A)

5.4 Deleting Cell Content

Action	Key / Method	Effect
Delete content only	Press Delete key	Removes the data but keeps the cell formatting intact
Delete content and formatting	Home tab > Editing group > Clear > Clear All	Removes data and all formatting from the cell
Delete formatting only	Home tab > Editing group > Clear > Clear Formats	Removes formatting but keeps the data
Delete the entire row or column	Right-click row/column header > Delete	Removes the row or column and shifts remaining data

📝 **Note — Delete vs. Backspace**

The Delete key removes the content of a selected cell. The Backspace key also removes content but puts the cell into edit mode. When multiple cells are selected, use Delete to clear them all at once.

Chapter 6

Customizing the Quick Access Toolbar

The Quick Access Toolbar (QAT) is a small, customizable bar that sits at the very top of the Excel window. It provides one-click access to the commands you use most frequently, regardless of which Ribbon tab you are on.

6.1 What is the Quick Access Toolbar?

Quick Access Toolbar (QAT)	A small, user-customizable toolbar located above the Ribbon (or optionally below it) that provides instant access to your most frequently used commands with a single click.

By default, the Quick Access Toolbar shows three commands: Save, Undo, and Redo. However, you can add any command you use regularly to save time.

6.2 Adding Commands to the Quick Access Toolbar

There are two easy ways to add commands to the Quick Access Toolbar:

Method 1 — Using the Dropdown Arrow

Step	Action
1	Click the small dropdown arrow (▼) at the right end of the Quick Access Toolbar
2	A list of popular commands appears (e.g., New, Open, Print Preview, Spelling)
3	Click the command you want to add. It will appear immediately in the toolbar

Method 2 — Right-Click any Ribbon Button

Find the command anywhere in the Ribbon, right-click on it, and select 'Add to Quick Access Toolbar'. The button will instantly appear in the QAT.

6.3 Removing Commands from the Quick Access Toolbar

To remove a command: Right-click the command button in the QAT > Select 'Remove from Quick Access Toolbar'.

💡 **Pro Tip — Customize for Your Role**

Add the commands you use every day to the QAT. For example:

- An accountant might add: AutoSum, Format as Table, Paste Special
- A teacher might add: Print, Spelling & Grammar, Save As
- A data analyst might add: Sort A-Z, Filter, Remove Duplicates

Chapter 7

Excel File Formats

When you save an Excel file, you choose a file format. Each format serves a different purpose. Understanding which format to use prevents common problems like losing formulas or being unable to share files with others.

7.1 Common Excel File Formats

Format	Extension	What It Supports	When to Use
Excel Workbook	.xlsx	All Excel features: formulas, charts, multiple sheets, formatting	Standard format for everyday work — use this by default
Excel Macro-Enabled Workbook	.xlsm	All features of .xlsx, plus VBA macros (automated scripts)	When the file contains or needs to run macros
Excel Binary Workbook	.xlsb	All features of .xlsx but stored in binary format (smaller file size)	For very large files where performance is a concern
Excel 97-2003 Workbook	.xls	Older Excel format with fewer features supported	Only when sharing with someone using Excel 2003 or older
CSV (Comma Separated Values)	.csv	Plain text only — no formulas, no formatting, no multiple sheets	For exporting data to other software, databases, or systems
PDF	.pdf	Fixed-layout document, not editable as a spreadsheet	When sharing a read-only, print-ready version of the file

🔎 **Analogy — File Formats Like Document Types**

Think of file formats like different types of packaging for the same content:

- .xlsx = A standard box with all your items inside, fully usable
- .csv = The items removed from the box and listed on plain paper (simple, text-only)
- .pdf = A photo of the box — you can see it, but you cannot change it
- .xlsm = A standard box with a built-in automated mechanism inside

7.2 Saving in a Different Format

To save a file in a different format: Click File > Save As > In the 'Save as type' dropdown, select the desired format > Click Save.

> 📝 **Important Note — Saving as CSV**
>
> If you save a file as .csv, Excel will warn you that only the current sheet will be saved, and all formatting and formulas will be lost. CSV saves only the raw data as plain text. Always keep a separate .xlsx copy of your file before saving as CSV.

Chapter 8

Basic Data Types in Excel

Excel handles different types of data differently. Understanding how Excel classifies and treats data is essential for avoiding errors in calculations and formatting. This chapter covers the three main data types in Excel: Numbers, Text, and Dates.

8.1 Numbers

Numbers	Numeric values that can be used in calculations. Excel stores numbers as values and aligns them to the right side of the cell by default.

Numbers in Excel include:

- Integers — whole numbers such as 10, 500, -25
- Decimals — numbers with a decimal point such as 3.14, 99.99, -0.5
- Negative numbers — numbers preceded by a minus sign such as -100
- Percentages — such as 15% (Excel stores this as 0.15 internally)
- Currency values — such as ₹1,500.00 or $250.00 (these are just numbers with a currency format applied)

📝 **Note — Numbers Stored as Text**

Sometimes numbers look correct but are stored as text (you may see a small green triangle in the corner of the cell). Text-numbers cannot be used in calculations. To fix this: Click the cell, look for the warning icon, and select 'Convert to Number'.

8.2 Text (Labels)

Text	Any data that is not treated as a number or date. Text is also called a Label. Excel aligns text to the left side of the cell by default.

Text in Excel is used for:

- Names — such as 'Alice Sharma', 'Mumbai', 'Marketing Department'
- Headings — such as column headers like 'Product Name', 'Invoice Number'
- IDs and codes — such as 'EMP-001', 'INV-2025-004' (these look like numbers but are labels)
- Descriptions — such as 'Pending', 'Approved', 'Out of Stock'

💡 **Tip — Forcing a Number to be Stored as Text**

Sometimes you need a number to be treated as text — for example, employee codes like '00042' (the leading zeros must not be dropped).

To do this: Type an apostrophe before the number: '00042

Excel will store it as text and display '00042' with the leading zeros intact.

The apostrophe itself is not displayed in the cell.

8.3 Dates and Times

Dates	Excel stores dates as serial numbers internally. For example, 1 January 1900 is serial number 1, and each subsequent day increments by 1. This allows Excel to perform date arithmetic such as calculating the number of days between two dates.

When you type a date into a cell, Excel recognizes it and formats it automatically. Common ways to type dates:

What You Type	How Excel Displays It	Notes
15/04/2025	15-04-2025 or April 15, 2025	Recognized as a date in DD/MM/YYYY format
April 15, 2025	15-Apr-25	Recognized as a date from text input
15-04-25	15-04-2025	Short year is interpreted as 2025
15/04/2025 09:30	15-04-2025 09:30	Recognized as both a date and a time

💡 **Pro Tip — Date Arithmetic**

Because dates are stored as numbers, you can subtract one date from another to find the number of days between them.

Example: If cell A1 = 01/04/2025 and cell B1 = 30/04/2025

Formula: =B1-A1 gives the result: 29 (days between the two dates)

Chapter 9

Basic Excel Operations — Copy, Cut, and Paste

Three of the most frequently used operations in Excel are Copy, Cut, and Paste. These operations allow you to duplicate data, move data, and insert it in different locations efficiently. This chapter covers all variations.

9.1 Copy, Cut, and Paste — Overview

Operation	What It Does	Keyboard Shortcut	Ribbon Location
Copy	Duplicates the selected cell(s). The original remains in place.	Ctrl + C	Home > Clipboard > Copy
Cut	Removes the selected cell(s) and places them in the clipboard. The original is deleted once you paste.	Ctrl + X	Home > Clipboard > Cut
Paste	Inserts the copied or cut content into the selected destination cell.	Ctrl + V	Home > Clipboard > Paste

9.2 Step-by-Step: Copying Data

Step	Action
1	Select the cell or range you want to copy (e.g., A1:A5)
2	Press Ctrl + C. A dashed animated border (marching ants) appears around the selection.
3	Click the destination cell where you want the copy to appear (e.g., C1)
4	Press Ctrl + V to paste. The data appears in the new location. The original data is still in A1:A5.

9.3 Paste Special

Standard paste (Ctrl + V) pastes everything: values, formulas, and formatting. Paste Special lets you choose exactly what to paste.

To access Paste Special: Press Ctrl + Alt + V, or click Home > Clipboard > Paste dropdown > Paste Special.

Paste Special Option	What It Pastes	Common Use Case

Values	Only the data result, not the formula	When you want a number, not a formula that might change
Formats	Only the formatting (colors, borders, fonts), not the data	Copying a style from one cell and applying it to another
Formulas	Only the formula, not the formatting	Copying a calculation to a new area
Transpose	Rotates the data: rows become columns and columns become rows	Converting a horizontal list into a vertical list
Column Widths	Applies the source column width to the destination	Making a new column the same width as an existing one

🔎 **Analogy — Paste Special is like Photocopying**

Imagine you have a printed receipt:

- Standard copy (Ctrl+V) = Photocopy the entire receipt exactly as it is
- Paste Values only = Type out just the total amount from the receipt on a new paper
- Paste Formats only = Apply the same paper color and font style to a blank receipt

Paste Special gives you control over exactly what you are duplicating.

Chapter 10

Autofill and Flash Fill

Excel includes two powerful features that help you fill in data quickly without typing every value manually. Autofill detects patterns in your data and extends them automatically. Flash Fill recognizes patterns in adjacent columns and fills the rest of your data accordingly.

10.1 Autofill

Autofill	A feature that automatically extends a pattern or series when you drag the Fill Handle across adjacent cells. It works for numbers, dates, days, months, and custom lists.

The Fill Handle is the small green square that appears at the bottom-right corner of a selected cell or range.

What You Enter	What Autofill Produces	Pattern Detected
1, 2	3, 4, 5, 6, 7...	Increment by 1
5, 10	15, 20, 25, 30...	Increment by 5
Monday	Tuesday, Wednesday, Thursday...	Days of the week
January	February, March, April...	Months of the year
01/04/2025	02/04/2025, 03/04/2025...	Date increment by 1 day
Q1	Q2, Q3, Q4	Quarter pattern
Product 1	Product 2, Product 3...	Text + number pattern

> 💡 **Pro Tip — Autofill Without Dragging**
>
> If you have data in the column next to your series, you can double-click the Fill Handle instead of dragging it. Excel will automatically fill down to match the length of the adjacent data column.

10.2 Flash Fill

Flash Fill	An intelligent Excel feature that automatically fills a column based on a pattern it detects from your example entries. It is especially useful for splitting, combining, or reformatting text data.

Flash Fill works when Excel recognizes what you are trying to do from the first one or two examples you type. It requires the data it learns from to be in an adjacent column.

Column A (Source Data)	Column B (Your Example)	Flash Fill Completes Column B As
Alice Sharma	Alice	Bob, Carol, David...
alice.sharma@mail.com	alice.sharma	bob.jones, carol.white...
9876543210	(987) 654-3210	(776) 543-2100, (665) 432-1099...
15/04/2025	April 2025	May 2025, June 2025...

To use Flash Fill: Type your example in the first cell of a new column. Press Ctrl + E to trigger Flash Fill, OR go to Data tab > Flash Fill.

Chapter 11

Sorting and Filtering Data

When working with lists of data in Excel, you often need to organize it in a specific order or narrow it down to see only relevant information. Sorting rearranges your data, while Filtering temporarily hides rows that do not meet your criteria.

11.1 Sorting Data

Sorting	Rearranging the rows of a dataset in a specific order based on the values in one or more columns. Sorting does not delete any data — it just reorders the rows.

Sort Type	Description	Example
Sort A to Z (Ascending)	Alphabetical order for text; Smallest to Largest for numbers; Oldest to Newest for dates	Names sorted A, B, C... or prices 100, 200, 300...
Sort Z to A (Descending)	Reverse alphabetical for text; Largest to Smallest for numbers; Newest to Oldest for dates	Prices sorted 1000, 500, 200...
Multi-level Sort	Sort by more than one column. The first column is sorted first, then ties are broken by the second column.	Sort by Department, then by Salary within each department

To sort your data: Click anywhere inside your data range. Go to Data tab > Sort A to Z or Sort Z to A. Alternatively, use the sort buttons in the Data tab for a custom multi-level sort.

11.2 Filtering Data

Filter	A tool that temporarily hides rows that do not meet the criteria you specify, showing only the rows that match. The hidden rows are not deleted — they are just out of view.

To apply a Filter: Click anywhere inside your data range. Go to Data tab > Filter. Small dropdown arrows will appear in the header row of each column. Click the dropdown arrow on the column you want to filter and select your criteria.

Filter Type	How to Apply	Example

Filter by specific value	Tick the checkboxes for the values you want to see	Show only rows where Department = 'Finance'
Number Filters	Dropdown > Number Filters > options like Greater Than, Between, Top 10	Show only rows where Sales > 50,000
Text Filters	Dropdown > Text Filters > options like Contains, Begins With, Ends With	Show only rows where Product Name contains 'Pro'
Date Filters	Dropdown > Date Filters > options like This Month, Last Year, Between	Show only orders from last month
Clear a filter	Click the dropdown arrow > Clear Filter From [Column Name]	Remove the Department filter to show all rows again

📝 **Note — How to Tell if a Filter is Active**

When a filter is active on a column, the dropdown arrow changes to a funnel icon (▼ becomes 🔽). Also, the row numbers of hidden rows are shown in blue instead of black, and some row numbers will appear to be skipped.

Chapter 12

Basic Cell Formatting — Font, Color, and Alignment

Formatting makes your spreadsheet easier to read, more professional, and visually organized. In this chapter, you will learn how to change fonts, apply colors, align text within cells, and use basic formatting tools from the Home tab.

12.1 Font Formatting

Font formatting changes how the text inside a cell looks. All font options are found in the Font group on the Home tab.

Formatting Option	Button in Ribbon	Keyboard Shortcut	What It Does
Bold	B	Ctrl + B	Makes text thicker and heavier for emphasis
Italic	I	Ctrl + I	Tilts text to the right for titles or notes
Underline	U	Ctrl + U	Adds an underline beneath text
Font Name	Dropdown showing font name	(none)	Changes the typeface of the text (e.g., Arial, Calibri, Times New Roman)
Font Size	Number box next to font name	Ctrl + Shift + > (increase)	Changes how large or small the text appears
Font Color	A with colored underline	(none)	Changes the color of the text
Fill Color	Paint bucket icon	(none)	Adds a background color to the cell

12.2 Alignment

Alignment controls where text is positioned within a cell. The Alignment group is next to the Font group on the Home tab.

Alignment Option	What It Does	Best Used For
Left Align	Text starts at the left edge of the cell	Text and labels (default for text)
Center Align	Text is centered in the cell	Column headers and short labels

Right Align	Text is pushed to the right edge of the cell	Numbers and currency (default for numbers)
Top Align	Content placed at the top of the cell	Cells with tall row height
Middle Align	Content is centered vertically in the cell	When cells have increased row height
Bottom Align	Content placed at the bottom of the cell	Default vertical position
Wrap Text	Long text wraps to a new line within the same cell	When a cell contains a long sentence or description
Merge & Center	Combines multiple cells into one and centers the content	For titles that span across multiple columns

📝 **Note — Merge & Center Caution**

While Merge & Center looks attractive for titles, it can cause problems when sorting or filtering data. Avoid merging cells inside data tables. Use it only for decorative headers above your data.

12.3 Number Formatting

Number formatting changes how numbers are displayed without changing their actual value. A cell showing ₹1,500.00 and a cell showing 1500 both store the same value; only the display is different.

Format Type	Example Display	How to Apply
General (default)	1500	No special format applied
Number	1,500.00	Home > Number group > Number format dropdown > Number
Currency (₹ / $)	₹1,500.00 or $1,500.00	Home > Number group > Currency button
Percentage (%)	15%	Home > Number group > Percentage button (%)
Date	15-Apr-25	Home > Number group > dropdown > Short Date or Long Date
Text	00042	Home > Number group > dropdown > Text (preserves leading zeros)
Accounting	₹ 1,500.00	Similar to currency but aligns symbols and decimals in a column

Chapter 13

Using the Clipboard

The Clipboard is a temporary storage area that holds items you have copied or cut. Excel has a built-in Clipboard panel that can hold up to 24 items at a time, allowing you to paste any of them whenever needed.

13.1 The Office Clipboard

Office Clipboard	A temporary storage area shared across Microsoft Office applications (Excel, Word, PowerPoint) that holds multiple copied or cut items. It allows you to paste any stored item, not just the most recently copied one.

The Office Clipboard is different from the standard Windows clipboard, which holds only one item at a time. The Office Clipboard can hold up to 24 items.

To open the Clipboard panel: Go to Home tab > Click the small arrow (↗) at the bottom-right corner of the Clipboard group. The Clipboard pane will appear on the left side of your screen.

Clipboard Action	How to Perform It
Open the Clipboard panel	Home > Clipboard group > click the arrow launcher (small diagonal arrow in the corner)
Paste a specific item	Click the cell where you want to paste, then click the item in the Clipboard panel
Paste all items at once	Click the 'Paste All' button at the top of the Clipboard panel
Delete a specific item	Hover over the item in the panel, click the dropdown arrow > Delete
Clear all items	Click the 'Clear All' button at the top of the Clipboard panel

🔎 Analogy — Clipboard Like a Shopping Cart

Imagine you are shopping online and you want to compare several items. You add them all to your cart first, then decide what to purchase.

The Office Clipboard works the same way: you copy multiple items during your work (they all go into the 'cart'), and then you paste whichever item you need, whenever you need it.

Chapter 14

Adjusting Column Width and Row Height

When you enter data in Excel, you will often find that the column is not wide enough to display all the text, or that rows look cramped. Adjusting column widths and row heights makes your spreadsheet clear, readable, and professional.

14.1 Why Adjust Column Width?

When a column is too narrow for its content:

- Numbers display as ###### (hash symbols) instead of the actual value
- Text is cut off and only the beginning is visible
- Dates may appear as numbers

14.2 Methods to Adjust Column Width

Method	How To Do It	Best Used For
Drag the border manually	Hover over the border between two column letters (e.g., between A and B) until you see the double-headed arrow (↔), then drag left or right	When you want a specific width by feel
AutoFit — double click	Hover over the right border of the column header until you see ↔, then double-click	Instantly fits the column to its widest content
AutoFit via menu	Select columns > Home > Cells > Format > AutoFit Column Width	For multiple columns at once
Set an exact width	Right-click the column header > Column Width > type a number > OK	When a specific width is required (e.g., for printing)
AutoFit multiple columns	Select multiple column headers, then double-click any border between selected columns	Fitting several columns in one action

14.3 Methods to Adjust Row Height

Row height adjustment works the same way as column width, but operates on horizontal row borders.

Method	How To Do It

Drag the border manually	Hover over the bottom border of a row number (e.g., below row 3) until you see ↕, then drag up or down
AutoFit row height	Select the row(s) > Home > Cells > Format > AutoFit Row Height
Set an exact height	Right-click the row number > Row Height > type a value > OK

💡 **Pro Tip — Select All Before AutoFit**

Press Ctrl + A to select all cells in the worksheet, then double-click any column border to AutoFit all columns at once. This is the fastest way to make an entire worksheet tidy and readable in one step.

Textbook Summary

Chapter Summary

- **Microsoft Excel** is a spreadsheet application used to organize, calculate, analyze, and visualize data in a grid of rows and columns.
- **The Excel Interface** consists of key components: Title Bar, Ribbon, Name Box, Formula Bar, Worksheet Grid, Sheet Tabs, and Status Bar.
- **The Ribbon** organizes all Excel commands into tabs (Home, Insert, Data, etc.) and groups (Font, Alignment, etc.) for easy access.
- **Workbooks** are Excel files (.xlsx) that contain one or more Worksheets — the individual grid pages of data.
- **Cells** are identified by their address (Column Letter + Row Number, e.g., B4) and can hold text, numbers, dates, or formulas.
- **Creating and Saving** workbooks uses standard shortcuts: Ctrl + N (New), Ctrl + S (Save), F12 (Save As). The default format is .xlsx.
- **File Formats** include .xlsx (standard), .xlsm (with macros), .csv (text-only), and .pdf (read-only print version).
- **Data Types in Excel** are Numbers (right-aligned, used in calculations), Text (left-aligned, labels), and Dates (stored as serial numbers internally).
- **Copy (Ctrl+C), Cut (Ctrl+X), Paste (Ctrl+V)** are core operations. Paste Special (Ctrl+Alt+V) lets you paste only values, formats, or other specific content.
- **Autofill** extends patterns (sequences, dates, weekdays) by dragging the Fill Handle. Flash Fill (Ctrl+E) automatically reformats text based on examples.
- **Sorting** rearranges data A-Z, Z-A, or by custom multiple levels. Filtering temporarily hides rows that do not match your criteria.
- **Cell Formatting** includes Font (bold, italic, color), Alignment (left, center, wrap), and Number format (currency, percentage, date).
- **The Clipboard** holds up to 24 copied items for reuse. Open it from Home > Clipboard group arrow launcher.
- **Column Width and Row Height** can be adjusted by dragging borders, using AutoFit, or entering exact values via right-click menus.